

2026 /03/15

1. Which of the following physical quantities is a scalar?
 - A. velocity
 - B. accelerated speed
 - C. displacement
 - D. distance travelled

2. The gravitational force acting on an object on the Moon is 30 N. Given that the Moon's gravitational acceleration is one-sixth of Earth's, what is the gravitational force acting on the same object on Earth?
 - A. 5 N
 - B. 30 N
 - C. 180 N
 - D. 300 N

3. A spring with an original length of 0.4 m and a stiffness coefficient $k = 15 \text{ N/m}$. When compressed by a force of 3 N, the new length of the spring is
 - A. 0.2 m
 - B. 0.4 m
 - C. 0.6 m
 - D. 0.8 m

4. When an object is subjected to two coplanar forces with magnitudes of 6 N and 8 N respectively, the resultant force may have a magnitude of
 - A. 0 N
 - B. 1 N
 - C. 10 N
 - D. 15 N

5. In a uniform magnetic field, a straight conductor with a length of 0.2 m is placed perpendicular to the magnetic field direction. When a current of 4 A flows through it, the resulting Ampere force is 0.8 N. Determine the magnetic induction intensity.
 - A. 0.1 T
 - B. 0.5 T
 - C. 1.0 T
 - D. 2.0 T

6. Which of the following statements about a point mass is correct?
 - A. When studying Earth's rotation, the Earth can be considered as a point mass.
 - B. When studying the time it takes for a train to pass through a tunnel, the train can be treated as a point mass.
 - C. When studying the high jump movement of athletes, the athlete can be considered as a point mass.
 - D. A point mass is an idealized model.

7. 摩天轮的半径为 R ，座舱随轮转动一周，其经过的路程和发生的位移分别为



A 路程 $2\pi R$ ，位移 $2R$

B 路程 $2R$ ，位移 $2\pi R$

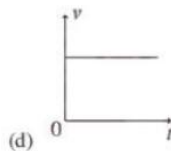
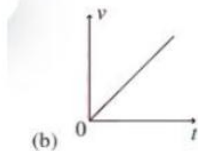
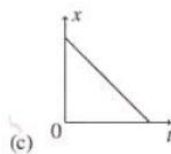
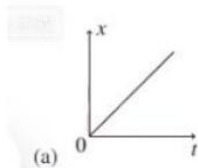
C 路程 $2R$ ，位移 0

☒ D 路程 $2\pi R$ ，位移 0

8. Which of the following statements is correct?

- A. The direction of sliding friction force is always opposite to the direction of object motion.
- B. The magnitude of static friction force is directly proportional to the normal pressure.
- C. The tighter the bottle is held, the greater the frictional force exerted on the bottle.
- D. The frictional force exerted on the feet during walking is static friction.

9. 下列哪个图像能表示汽车在平直路面上做匀加速直线运动的状态



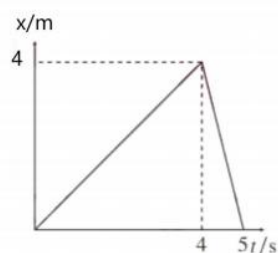
10. An object falls freely from a height of 80 m with $g = 10\text{ m/s}^2$. The time required to reach the ground is

- A. 2 s
- B. 4 s
- C. 6 s
- D. 8 s

11. An object undergoing uniform circular motion has a radius of 0.5 m and a period of 2 s . Its linear velocity is

- A. $0.5\pi\text{ m/s}$
- B. $\pi\text{ m/s}$
- C. $1.5\pi\text{ m/s}$
- D. $2\pi\text{ m/s}$

12. 某玩具小车做直线运动的位移 - 时间 ($x-t$) 图像如图所示。下列说法正确的是



- A 0-4 秒内, 玩具小车的位移为 8 米 B 0-4 秒内, 玩具小车的速度为 1.0 米 / 秒
 C 0-5 秒内, 玩具小车的平均速度为 0.8 米 / 秒 D 4-5 秒内, 玩具小车的速度为 0.8 米 / 秒

13. The elevator decelerates downward with an acceleration of 1 m/s^2 , and the pressure exerted by a person with a mass of 50 kg on the elevator floor is ($g = 10 \text{ m/s}^2$).

- A. 450 N
 B. 500 N
 C. 550 N
 D. 600 N

14. For an object with a mass of 5 kg sliding on a horizontal ground with a sliding friction force of 10 N, the dynamic friction coefficient is ($g = 10 \text{ m/s}^2$)

- A. 0.1
 B. 0.2
 C. 0.3
 D. 0.4

15. On a smooth horizontal surface, an object with a mass of 3 kg is subjected to a constant force, and its velocity increases from 2 m/s to 8 m/s within 2 seconds. The magnitude of the force is

- A. 3 N
 B. 6 N
 C. 9 N
 D. 12 N

16 光从空气射入某介质时, 入射角 $i=60^\circ$, 折射角 $r=30^\circ$ 。已知空气的折射率 $n=1$, 则该介质的折射率为

A $\sqrt{3}$

B $\frac{\sqrt{3}}{3}$

C 1.5

D $\sqrt{5}$

17. The small sphere undergoes uniform circular motion with an angular velocity of $200\pi \text{ rad/s}$ and a radius of 0.05 m, resulting in a centripetal acceleration of magnitude

- A. $2000\pi^2 \text{ m/s}^2$
 B. $1000\pi^2 \text{ m/s}^2$
 C. $200\pi^2 \text{ m/s}^2$
 D. $20\pi^2 \text{ m/s}^2$

18. The spring oscillator performs simple harmonic motion with an amplitude of 5 cm and a period of 4s. Starting the timing from the equilibrium position, the distance traveled within 2s is
- A. 5 cm
 - B. 10 cm
 - C. 15 cm
 - D. 20 cm
19. An object with a mass of 3 kg moving at a velocity of 4 m/s has a momentum magnitude of
- A. 7 kg·m/s
 - B. 12 kg·m/s
 - C. 24 kg·m/s
 - D. 48 kg·m/s
20. A water drop falls freely from a height of 20 m without air resistance. The landing speed is ($g = 10 \text{ m/s}^2$).
- A. 10 m/s
 - B. 15 m/s
 - C. 20 m/s
 - D. 25 m/s
21. When a horizontal object is pulled forward by a constant horizontal force F over a distance of 5 m, doing work of 50 J, the magnitude of F is
- A. 5 N
 - B. 10 N
 - C. 15 N
 - D. 20 N
22. For a simple harmonic transverse wave with a wavelength of 8 m and a period of 2s, the wave velocity is
- A. 2 m/s
 - B. 4 m/s
 - C. 8 m/s
 - D. 16 m/s
23. Two resistors with resistance values of 2Ω and 8Ω , respectively, have a total parallel resistance of
- A. 1.6Ω
 - B. 2Ω
 - C. 4Ω
 - D. 10Ω
24. In the circuit diagram shown, $R = 4 \Omega$. When S is open, the voltage reading is 6 V; when S is closed, the reading is 4 V. Therefore, the internal resistance of the power supply is
- A. 0.5Ω
 - B. 1Ω
 - C. 2Ω
 - D. 3Ω
25. For an ideal gas of a given mass, if the pressure remains constant and the volume increases, then the gas's

- A. temperature increment
- B. fall of temperature
- C. conservation of internal energy
- D. exothermic release

26. When an object is vertically thrown from the ground, the work done by gravity during its ascent is -100 J , then

- A. Kinetic energy decreased by 100 J
- B. The gravitational potential energy decreased by 100 J
- C. Mechanical energy decreased by 100 J
- D. Kinetic energy increased by 100 J

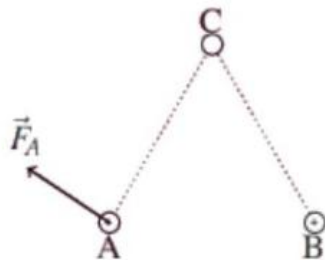
27. The current flowing through the conductor is 0.8 A , so the charge passing through the conductor's cross-section within 2 seconds is

- A. 0.4 C
- B. 1.6 C
- C. 2.4 C
- D. 3.2 C

28 一定质量的理想气体，温度保持不变，压强减小，则气体的体积会

- A 以上情况都有可能
- B 增大
- C 减小
- D 保持不变

29 三个点电荷 A、B、C 位于等边三角形的三个顶点，其中 A、B 均带正电。B、C 两个电荷对 A 的静电力的合力为 F_A ，如图所示。已知 F_A 与 BA 的延长线夹角小于 60° ，则关于电荷 C 的判断正确的是



- A 带正电，电荷量小于 B 的电荷量
- B 带负电，电荷量大于 B 的电荷量
- C 带正电，电荷量大于 B 的电荷量
- D 带负电，电荷量小于 B 的电荷量

30. In a uniform electric field, which of the following statements is correct?

- A. The potentials at all points are equal
- B. The field strengths at all points are equal.
- C. The electric field lines are curves
- D. The direction of the electric field varies everywhere.

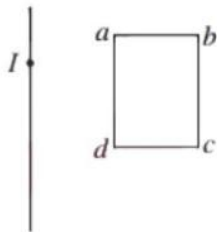
31. The wire is 0.5 m long, with a current of 3 A and a magnetic field of 0.2 T. The direction of the magnetic field forms a 60° angle with the current. The magnitude of the Ampere force is

- A. 0.15 N
- B. 0.3 N
- C. 0.45 N
- D. 0.6 N

32. The coil area is 0.3 m^2 , the magnetic field is 0.5 T, the angle between the normal and the magnetic field direction is 30° , and the magnetic flux is

- A. 0.075 Wb
- B. 0.15 Wb
- C. 0.3 Wb
- D. 0.45 Wb

33 如图所示，一根通有电流 I 的长直导线右侧，固定一个矩形金属线框 $abcd$ ，线框的 cd 边与导线平行。调节电流 I ，使空间各点的磁感应强度随时间均匀增大，则



- A 线框中的感应电流方向为 $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$
- B 线框中的感应电流逐渐增大
- C 线框受到的安培力方向水平向右
- D 线框 cd 边受到的安培力大小保持不变

34 如图所示，空间存在电场， a 、 b 为电场中的两点。下列说法正确的是。



- A b 点的电场强度大于 a 点的电场强度
- B a 点的电场强度大于 b 点的电场强度
- C 该电场为匀强电场
- D a 、 b 两点的电场强度大小相等

35. When a beam of light travels from air to glass at an incident angle of 45° and a refractive angle of 30° , the refractive index of the glass is

- A. $\sqrt{2}$
- B. $\sqrt{3}$
- C. 1.5
- D. 2

37. An object of mass m is thrown vertically upward from the ground with v_0 . Ignoring air resistance, the mechanical energy at the highest point is (with the ground as the zero potential energy surface).

- A. $\frac{1}{2}mv_0^2$
- B. mgh
- C. $\frac{1}{2}mv_0^2 + mgh$
- D. 0

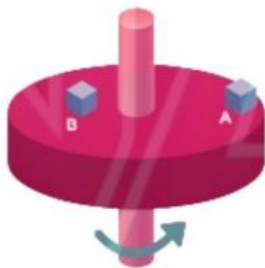
38. The vehicle has a mass of 1.5 t and a power output of 45 kW. When traveling at a constant speed of 54 km/h, the resistance is

- A. 1000 N
- B. 1500 N
- C. 2000 N
- D. 3000 N

39. If the initial velocity of projectile motion increases to three times its original value, then

- A. The flight time has increased to three times the original duration.
- B. The horizontal displacement has tripled from the original value.
- C. landing speed remains constant
- D. Vertical displacement increases to three times the original value

40 如图所示，两个质量相等的物体 A、B 放在绕竖直轴匀速转动的水平圆盘上，且两物体相对圆盘保持静止，则



- | | |
|---------------------------|---------------------------|
| A 物体 A 的角速度与物体 B 的角速度大小相等 | B 物体 A 的线速度与物体 B 的线速度大小相等 |
| C 物体 A 的角速度大于物体 B 的角速度 | D 物体 A 的角速度小于物体 B 的角速度 |

41. The power of gravitational work done during the second second of free-fall motion ($g = 10 \text{ m/s}^2$, $m = 1 \text{ kg}$) is

- A. 50 W
- B. 100 W
- C. 150 W
- D. 200 W

42. The metal rod moves uniformly to the right along the guide rail, passing through the magnetic flux of the closed loop.

- A. magnify
- B. abate
- C. invariant
- D. Increase first, then decrease

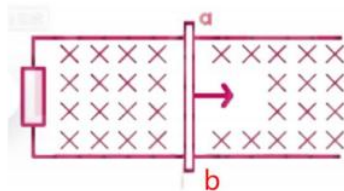
43. Two charges, $+q$ and $-2q$, are separated by a distance L . A charge q is placed on the extension line. After contact, the charges are released. The ratio of electrostatic forces is

- A. 2:1
- B. 1:2
- C. 9:4
- D. 4:9

44. A ball with a mass of 0.2 kg collides with a wall at 5 m/s , rebounds at 3 m/s , and has a contact time of 0.05 s . The average force is

- A. 8 N
- B. 16 N
- C. 24 N
- D. 32 N

46 如图所示，水平放置的平行金属导轨两端连接有电阻 R ，导线 ab 可在导轨上无摩擦滑动，匀强磁场垂直穿过导轨平面。当导线 ab 向右匀速滑动时，下列说法正确的是



- A 只要导线 ab 达到某一速度，撤去外力后，导线 ab 也能在导轨上保持匀速运动
- B 导线 ab 既受到拉力作用，又受到安培力作用
- C 导线 ab 的运动速度不变时，增大电阻 R 的阻值，拉动导线 ab 的拉力大小可以增大
- D 导线 ab 的运动速度越大，所需的拉力大小越小

48. The maximum acceleration of the vehicle is 4 m/s^2 , the speed is 72 km/h , the reaction time is 1 s , and the minimum safe distance is

- A. 20 m
- B. 30 m
- C. 40 m
- D. 50 m